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Welding transformer

Abstract

A flat, electrically and thermally conductive flat surface group is formed in the common connection region **32** of the secondary coil **13**. The flat conductive surface group is directly and mechanically joined to the connecting surfaces of the first conductor plate **42**, the second conductor plate **44**, and the third conductor plate **46**, respectively. The first conductor plate **42**, the second conductor plate **44**, and the third conductor plate **46** cover the entire common connection region **32**. The annular first conductor plate **42** occupies the maximum area. The refrigerant can be circulated in the annular cavity provided inside the first conductor plate **42** to efficiently cool the whole.

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Field of Classification Search

CPC: H01F (27/10); H01F (38/085); H01F (27/08); H01F (2027/408); H01F (27/38); H01F (30/10); B23K (11/241); F28F (3/048)

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a welding transformer.

BACKGROUND OF THE INVENTION

(2) The inventors of this invention have developed a high-speed and high-quality resistance welding transformer and welding device. It controls the current of a primary coil by an inverter. Cooling water circulates in a secondary coil (Patent Document 1).

PRIOR ART DOCUMENT

(3) [Patent Document]

(4) [Patent Document 1] PCT/JP2012/066646

SUMMARY OF THE INVENTION

Problem to be Resolved by the Invention

(5) In the welding transformer introduced in Patent Document 1, cooling water is circulated in the secondary coils. The secondary coils are obtained cutting copper plates. The secondary coils are electrically connected together and have refrigerant passages. Due to this structure, it is compact and exhibits the performance of being able to output a large current. Features such as the ability to connect the output terminals of welding transformers in parallel and use them have been highly evaluated. While maintaining this performance, it was required to further simplify the structure and reduce the manufacturing cost. In order to solve this problem, the present invention provides a welding transformer described below.

Means of Solving of the Problems

(6) The following Configurations are structures for solving the above problems.

(7) <Configuration 1>

(8) A welding transformer, comprising: a primary coil (12) and a secondary coil (13) are wound on a magnetic core (17), the secondary coil (13) includes a positive side coil (14) and a negative side coil (16) being connected in series, one terminal of the positive side coil (14) and one terminal of the negative side coil (16) are electrically connected to a first common electrode (22) together, the other terminal of the positive side coil (14) is electrically connected to one terminal of a first rectifier element (18), the other terminal of the negative side coil (16) is electrically connected to one terminal of a second rectifier element (20), the other terminal of the first rectifier element (18) and the other terminal of the second rectifier element (20) are electrically connected to a second common electrode (24) together, the first common electrode (22) and the second common electrode (24) are for electrically connection to a welding apparatus (28), the primary coil (12) is for receiving primary electric current from an inverter, which invert pulse that reverses polarity at a predetermined repetition rate, the primary coil (12) is divided into plural parts, and wound on the magnetic core (17), the plural positive side coils (14) and the plural negative side coils (16) are inserted between the plural parts, electrically and thermally conductive flat surface (30a) (30b) (30c) is formed on each terminal of all positive coils (14) and all negative coils (16), all the conductive flat surfaces are arranged in a common connection region (32) with a minute interval (34), the conductive flat surfaces (30a) being formed on one terminal of the positive coils (14) and one terminal of the negative coils (16) are a first conductive surfaces, the conductive flat surfaces

(30b) being formed on the other terminal of the positive coils (14) are a second conductive surfaces, the conductive flat surfaces (30c) being formed on the other terminal of the negative coils (16) are a third conductive surfaces, the first conductive surfaces are directly and mechanically connected to the connection surface of a first conductor plate (42), the second conductive surfaces are directly and mechanically connected to the connection surface of the second conductor plate (44), the third conductive surfaces are directly and mechanically connected to the connection surface of the third conductor plate (46), the connection surface of the first conductor plate (42), the connection surface of the second conductor plate (44) and the connection surface of the third conductor plate (46) cover the whole common connection region (32), and the area of the connection surface of the first conductor plate (42) is the largest, a cavity (48) for flowing a cooling medium is provided inside the first conductor plate (42), the second conductor plates (44) are contacted to the one terminal of the first rectifier element (18) for electrically and thermally connection, and the third conductive plates (46) are contacted to the one terminal of the second rectifier element (20) for electrically and thermally connection.

<Configuration 2>

(9) The welding transformer according to the configuration 1, wherein the first conductor plate includes an annular portion that covers and contacts a peripheral portion of the common connection region, and a cavity (48) for circulating the cooling medium is provided inside the annular portion (50).

<Configuration 3>

(10) The welding transformer according to the configuration 2, wherein a connection surface of the second conductor (44) and a connection surface of the third conductor (46) are placed in a region surrounded by annular portion (50) of the first conductor plate (42).

<Configuration 4> wherein the connection surface of the first conductor plates (42) is formed with an uneven surface (56) include convex portions (52) and concave portions (54), that surfaces is in close contact with the first conductive surfaces on the common connection region (32).

<Configuration 5> wherein the connection surface of the second conductor plates (44) and the third conductor plates (46) is formed with an uneven surface (56) include convex portions (52) and concave portions (54), that surfaces is in close contact with the second conductive surfaces and the third conductive surfaces on the common connection region (32).

<Configuration 6>

(11) The welding transformer according to the configuration 2, Wherein the second conductor plate (44) and the third conductor plate (46) are provided with a cavity (50) for circulating the cooling medium inside the second conductor plate (44) and the third conductor plate (46).

Effect of this Invention

(12) A flat, electrically and thermally conductive flat surface group is formed in the common connection region 32 of the secondary coil 13. The conductive surface group is directly and mechanically connected to the connecting surfaces of the first conductor plate 42, the second conductor plate 44, and the third conductor plate 46, respectively. The first conductor plate 42, the second conductor plate 44, and the third conductor plate 46 cover the entire common connection region 32.

(13) Refrigerant can be circulated in the cavity of the first conductor plate 42 to efficiently cool the whole. Furthermore, the structure is simplified and the mechanical strength is high. When the annular portion 50 is provided on the first conductor plate 42, the refrigerant can be circulated in the annular cavity provided inside, and the whole can be efficiently cooled.

(14) If the second conductor plate 44 and the third conductor plate 46 are surrounded by the annular portion 50 of the first conductor plate 42, these can be efficiently cooled. When the conductive flat surface group and the conductor plate are joined via the uneven surface 56, the area for heat transfer is wide and the cooling performance is improved. Positioning can be performed with high accuracy by using the uneven surface 56, and the mechanical strength is increased.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an equivalent circuit of a welding transformer **10**.
- (2) FIG. 2 is a current waveform diagram showing the relationship between the switching current and the welding current.
- (3) FIG. 3 is an exploded perspective view of the welding transformer **10** of the present invention.
- (4) FIG. 4 is a perspective view of a secondary coil **13**.
- (5) FIG. 5A and FIG. 5B is a perspective view of a unit in which integrate a pair of positive side coils **14** and negative side coils **16**, and FIG. 5C is a partial side view showing a joined state of a conductor surface **30a** and a first conductor plate **42**.
- (6) FIG. 6 is a perspective view of a first conductor plate **42**.
- (7) FIG. 7 is a perspective view of a second conductor plate **44**.
- (8) FIG. 8 is a perspective view of a third conductor plate **46**.
- (9) FIG. 9 is a perspective view showing a state in which a first conductor plate **42**, a second conductor plate **44**, and a third conductor plate **46** are connected to a secondary coil **13**.
- (10) FIG. 10 is an external perspective view of a welding transformer **10** in which a main part is completed.

DESCRIPTION OF THE EMBODIMENT

- (11) Hereinafter, embodiments of the present invention will be described in detail for each example.

Embodiment 1

- (12) FIG. 1 is an equivalent circuit of the welding transformer **10**. The welding transformer **10** is a device for supplying a welding current to the welding machine **28**. The welding transformer **10** incorporates a primary coil **12**, a secondary coil **13**, a first rectifying element **18**, and a second rectifying element **20**.
- (13) The primary coil **12** and the secondary coil **13** of the welding transformer **10** are wound on a magnetic core **17**. The secondary coil **13** is a coil in which the positive coil **14** and the negative coil **16** are connected in series. A plurality of sets (7 sets in this embodiment) of the secondary coils **13** are electrically connected together in parallel to the secondary side of the welding transformer **10**.
- (14) One terminal of the positive coil **14** and one terminal of the negative coil **16** are electrically connected together to the first common electrode **22**. One terminal of the first rectifying element **18** is electrically connected to the other terminal of the positive coil **14**, and one terminal of the second rectifying element **20** is electrically connected to the other terminal of the negative coil **16**. The other terminal of the first rectifying element **18** and the other terminal of the second rectifying element **20** are electrically connected together to the second common electrode **24**.
- (15) The first common electrode **22** and the second common electrode **24** are electrically connected together to the welding machine **28**. A pulse-shaped primary current that inverts the polarity at a repetition frequency at a predetermined repetition rate by an inverter is supplied to the primary coil **12**.
- (16) FIG. 2 is a current waveform diagram showing the relationship between the switching current and the welding current. This graph shows the change in welding current from the start of one spot welding to the end of welding. When the switching voltage shown in FIG. 2 (a) is supplied to the primary coil **12**, the primary current shown in FIG. 2 (b) flows through the primary coil **12**. This current is full-wave rectified on the secondary side, and the welding electric current shown in FIG. 2 (c) is output from the welding transformer **10** and supplied to the welding machine **28**. (Overall structure)
- (17) FIG. 3 is an exploded perspective view of the welding transformer **10** of the present invention. The primary coil **12** is divided and wound into a plurality of parts and wound on a magnetic core

17. The structure of the primary coil **12** is the same as that introduced in Patent Document 1. The secondary coil **13** is sandwiched between the gaps of the split wound primary coil **12**. This structure is also the same as that introduced in Patent Document 1. A magnetic core **17** is inserted into the central portion of the primary coil **12** and the secondary coil **13**. The magnetic core **17a** is connected to the end of the magnetic core **17** in order to form a magnetic path in a loop shape.

(18) The first conductor plate **42** is connected to the secondary coil **13**. The first conductor plate **42** is for electrically connecting the first common electrode **22** and the secondary coil **13** described in FIG. The second conductor plate **44** and the third conductor plate **46** are inserted into holes provided in the first conductor plate **42** with an insulating sheet (not shown) sandwiched between them, and are connected to the secondary coil **13**.

(19) An insulating sheet (not shown) is sandwiched between the first conductor plate **42**, the second conductor plate **44**, and the third conductor plate **46** in order to electrically separate them from each other. The second conductor plate **44** is for electrically connecting the secondary coil **13** and the first rectifying element **18**.

(20) The third conductor plate **46** is for electrically connecting the secondary coil **13** and the second rectifying element **20**. The fourth conductor plate **47** and the first conductor plate **42** are connected in a U shape and surround the primary coil **12** and the secondary coil **13**.

(21) The first rectifying element **18** is sandwiched between the second conductor plate **44** and the first electrode plate **58**. The second rectifying element **20** is sandwiched between the third conductor plate **46** and the second electrode plate **60**. The first electrode plate **58** and the second electrode plate **60** are connected by a connecting plate **62**. The connecting plate **62** is for electrically connecting the first rectifying element **18** and the second rectifying element **20** to the second common electrode **24** described in FIG.

(22) The secondary coil **13**, the first conductor plate **42**, the second conductor plate **44**, the third conductor plate **46**, the fourth conductor plate **47**, the first electrode plate **58**, the second electrode plate **60**, and the connecting plate **62** are Both can be made by cutting a copper plate.

(23) (Structure of Secondary Coil)

(24) FIG. 4 is a perspective view of the secondary coil **13**. FIG. 5 is a perspective view showing a unit in which a pair of positive side coils **14** and a negative side coil **16** are integrated, and their conduction surfaces **30a**, **30b**, and **30c**. The positive side coil **14** and the negative side coil **16** constituting the secondary coil **13** can be manufactured by, for example, combining parts obtained by cutting copper into a shape as shown in the figure. One of the pair of one-turn coils arranged side by side is the positive coil **14**, and the other is the negative coil **16**. Each unit is mechanically connected by brazing or welding.

(25) (Conduction Surface)

(26) One terminal of the positive coil **14** and one terminal of the negative coil **16** are continuously integrated through a portion provided with a conductive flat surface **30a**. The conductive flat surface **30b** at the other terminal of the positive coil **14** projects from the central portion of the conductive flat surface **30a**. The conductive surface **30c** at the other terminal of the negative coil **16** also protrudes from the central portion of the conductive flat surface **30a**.

(27) As shown in FIG. 4, there are those in which the conductive surface **30a** has no unevenness as a whole and those in which both ends are one step higher than the central portion. In the embodiment of FIG. 4, three sets of each of the two types of units (shown in FIG. 5A and FIG. 5B) and one set of the pair of positive coils **14** and the negative coil **16** at the end are combined and integrated.

(28) In any of the units shown in FIG. 5A and FIG. 5B, one terminal portion **14a** and **16a** (FIG. 5A) of the positive coil **14** and the negative coil **16** are connected to the conductor surface **30a**. The other terminal portion **14b** (FIG. 5A) of the positive coil **14** is connected to the conductor surface **30b**. The other terminal portion **14c** (FIG. 5A) of the negative coil **16** is connected to the conductor surface **30c**.

(29) In the conduction surface **30a**, high and low height portions are alternately arranged in the plane of the common connection region **32** shown in FIG. **4**. The conductive surface **30b** and the conductive surface **30c** are arranged in a row at intervals. The surfaces of the conductive surface **30a**, the conductive surface **30b**, and the conductive surface **30c**, which are in contact with the first conductor plate **42**, the second conductor plate **44**, and the third conductor plate **46**, are flat and sufficiently wide to ensure electrical and thermal conduction. Conduction can be ensured.

(30) FIG. 5C is a partial side view showing a joint state between the conductor surface **30a** and the first conductor plate **42** as viewed from the side of the negative coil **16**. In this way, the conductor surfaces **30a** having different heights are arranged alternately, and the uneven surfaces **56** of the first conductor plate **42** described in FIG. **6** are in close contact with them without any gaps. The side surface on which the positive coils **14** on the opposite side are lined up has the same structure. Since the first conductor plate **42**, the positive coil **14** and the negative coil **16** are in contact with each other over a wide area, a high cooling effect can be obtained.

(31) The group of conductive surfaces **30a** formed at one terminal of all the positive coils **14** and one terminal of all the negative coils **16** is called a first conductive surface group. The group of conductive surfaces **30b** formed at the other terminal of all the positive coils **14** is called a second conductive surface group. The group of conductive surfaces **30c** formed at the other terminal of all the negative coils **16** is called a third conductive surface group.

(32) The first conductive surface group is mechanically directly and mechanically joined to the connecting surface of the first conductor plate **42**. The second conductive surface group is mechanically directly and mechanically joined to the connecting surface of the second conductor plate **44**. The third conductive surface group is directly mechanically joined to the connecting surface of the third conductor plate **46**.

(33) A gap **34** for electrical insulation is provided so that the conductive surface **30a**, the conductive surface **30b**, and the conductive surface **30c** are electrically insulated from each other, and for example, an insulating sheet (not shown) is sandwiched therein. The structure of the unit shown in FIG. **5** can be freely deformed so as to have the same function.

(34) (Common Connection Area)

(35) Conductive surfaces **30a** are arranged in a ring shape in the common connection area **32** so as to surround the entire surface of the common connection area **32**. Further, the conductive surface **30b** and the conductive surface **30c** are arranged in a row at a place surrounded by the conductive surface **30a**, respectively.

(36) (A First Conductor Plate)

(37) FIG. **6** is a perspective view of the first conductor plate **42**. The first conductor plate **42** is entirely composed of an L-shaped copper plate. An uneven surface **56** in which convex portions **52** and concave portions **54** are alternately provided is formed on the connecting surface of the annular portion **50**. As described in FIG. 5C, the uneven surface **56** is formed so as to be in close contact with all the conductive surfaces **30a** and the upper surface and the side surface thereof.

(38) As a result, the first conductor plate **42** is electrically and mechanically connected to the conduction surface **30a** provided at one terminal of all the positive coil **14** and the negative coil **16**. When the uneven surface **56** is brought into close contact with the upper surface and the side surface of the conductive surface **30a**, the surface area for heat transfer can be increased as compared with the case where the entire surface is in contact with the flat surface, so that the cooling efficiency is improved. Mechanical strength also increases. Further, if the uneven surface **56** is provided, the first conductor plate **42** can be accurately positioned with respect to the secondary coil **13**.

(39) In this embodiment, a cavity through which the cooling medium flows can be provided inside the fourth conductor plate **47** shown in FIGS. **3** and **10**. The fourth conductor plate **47** is arranged in noncontact with the primary coil **12** and the secondary coil **13** or via an insulator. The fourth conductor plate **47** has a function of indirectly cooling the primary coil **12** and the secondary coil

13.

(40) The annular portion **50** covers and contacts the peripheral portion of the surface of the common connection region **32**. Since the cavity for flowing a cooling medium **48** having a wide cross-sectional area can be formed in the first conductor plate **42**, a large amount of cooling medium such as cooling water can be circulated quickly. Therefore, the annular portion **50** can efficiently cool the secondary coil **13**. The shape, structure, and path of the cavity **48** can be freely selected in addition to this embodiment.

(41) The electrically and thermal conductive surface is a connecting surface on which a sufficient current necessary for the welding transformer **10** to operate normally can flow. The electrically and thermal conductive surface has a sufficiently wide surface for heat transfer. For example, the first conductive surface group is mechanically connected to the connection surface of the first conductor plate **42** by, for example, welding or brazing.

(42) (Second Conductor Plate)

(43) FIG. **7** is a perspective view of the second conductor plate **44**. An uneven surface **56** in which convex portions **52** and concave portions **54** are alternately arranged is formed in a portion of the second conductor plate **44** that is in close contact with the secondary coil **13**. The uneven surface **56** is in close contact with the upper surface and the side surface of the conductive surfaces **30b** arranged in a row shown in FIG. **4** and is electrically and mechanically connected.

(44) In this way, the uneven surface **56** and the conductive surface **30b** can be connected via a sufficiently large area. Further, by providing the uneven surface **56**, the second conductor plate **44** can be positioned with high accuracy and the connection work can be performed. The second conductor plate **44** is also provided with a cavity **48** (not shown) for circulating the cooling medium inside, and the cooling medium is supplied from the introduction port **68**, which is not shown on the opposite side. It can be discharged from the outlet port **66** and cooled.

(45) (Third Conductor Plate)

(46) FIG. **8** is a perspective view of the third conductor plate **46**. An uneven surface **56** in which convex portions **52** and concave portions **54** are alternately arranged is also formed in a portion of the third conductor plate **46** that is in close contact with the secondary coil **13**. The uneven surface **56** is in close contact with the upper surface and the side surface of the conductive surfaces **30c** arranged in a row shown in FIG. **4** and is electrically and mechanically connected. The effect of the uneven surface **56** is the same as that of the second conductor plate **44**. The structure capable of cooling is the same as that of the second conductor plate **44**.

(47) The connecting surface of the second conductor plate **44** and the third conductor plate **46** is placed in a region surrounded by the annular portion **50** of the first conductor plate **42**. The first conductor plate **42**, the second conductor plate **44**, and the third conductor plate **46** cover the entire common connection region **32**, and the first conductor plate **42** occupies the maximum area. This ensures high cooling capacity. (assembly) FIG. **9** shows a state in which the first conductor plate **42** is connected to the secondary coil **13** and the second conductor plate **44** and the third conductor plate **46** are fitted into the first conductor plate **42**. As described above, an insulating sheet (not shown) is sandwiched between the second conductor plate **44** and the third conductor plate **46**. Further, an insulating sheet is also sandwiched between the surfaces of the second conductor plate **44** and the third conductor plate **46** and the first conductor plate **42** in contact with each other.

(48) (Overall Appearance)

(49) FIG. **10** is an external perspective view of a welding transformer in which the main part is completed.

(50) The secondary coil **13** is sandwiched between the gaps of the separately wound primary coil **12**. A magnetic core **17** is inserted in the central portion of the primary coil **12** and the secondary coil **13**.

(51) The first rectifying element **18** (hidden in this figure) is fitted in the second conductor plate **44**. The second rectifying element **18** is sandwiched between the second conductor plate **44** and the

first electrode plate **58**. A second rectifying element **20** (hidden in this figure) is fitted in the third conductor plate **46**. The second rectifying element **20** is sandwiched between the third conductor plate **46** and the second electrode plate **60**.

(52) The terminal plate **70** and the terminal plate **72** are input terminals of the primary coil **12**. The first conductor plate **42** is connected to the first common electrode **22**. The connecting plate **62** is connected to the second common electrode **24**. The welding transformer **10** operates in this state. It is also possible to use a plurality of welding transformers **10** connected in parallel.

(53) As explained above, since the first conductor plate **42** has a very simple structure, a large cavity for flowing a cooling medium can be provided inside. Since the cooling medium can flow so as to circulate in the annular portion **50**, a smooth loop-like flow of the cooling medium can be realized. Since a large amount of refrigerant can be circulated in a short time, high cooling efficiency can be realized.

(54) Since the first conductor plate **42**, the second conductor plate **44**, and the third conductor plate **46** are in direct contact with the secondary coil **13** over a wide area, the heat transfer efficiency is good. The connection part has little electrical resistance loss.

REFERENCE SIGNS LIST

(55) **10** A welding transformer **12** primary coils **13** secondary coils **14** a positive side coil **16** a negative side coil **17** a magnetic core **18** a first rectifier element **20** a second rectifier element **22** a first common electrode **24** a second common electrode **28** welding devices **30a** conductive flat surface **30b** conductive flat surface **30c** conductive flat surface **32a** common connection region **34** Spacing for electrical insulation **42** a first conductor plate **44** a second conductor plate **46** a third conductor plate **47** a fourth conductor plate **48** a cavity for flowing a cooling medium **50** an annular portion **52** convex portions **54** concave portions **56** an uneven surface **58** first electrode plate **60** second electrode plate **62** connecting plate **64** introduction port **66** outlet port **68** introduction port **70** terminal plate **72** terminal plate

Claims

1. A welding transformer, comprising: a primary coil and a secondary coil are wound on a magnetic core, the secondary coil includes a positive side coil and a negative side coil being connected in series, one terminal of the positive side coil and one terminal of the negative side coil are electrically connected to a first common electrode together, the other terminal of the positive side coil is electrically connected to one terminal of a first rectifier element, the other terminal of the negative side coil is electrically connected to one terminal of a second rectifier element, the other terminal of the first rectifier element and the other terminal of the second rectifier element are electrically connected to a second common electrode together, the first common electrode and the second common electrode are for electrically connection to a welding apparatus, the primary coil is for receiving primary electric current from an inverter, which invert pulse that reverses polarity at a predetermined repetition rate, the primary coil is divided into plural parts, and wound on the magnetic core, the plural positive side coils and the plural negative side coils are inserted between the plural parts, electrically and thermally conductive flat surface is formed on each terminal of all positive coils and all negative coils, all the conductive flat surfaces are arranged in a common connection region with a minute interval, the conductive flat surfaces being formed on one terminal of the positive coils and one terminal of the negative coils are the first conductive surfaces, the conductive flat surfaces being formed on the other terminal of the positive coils are the second conductive surfaces, the conductive flat surfaces being formed on the other terminal of the negative coils are the third conductive surfaces, the first conductive surfaces are directly and mechanically connected to the connection surface of the first conductor plate, the second conductive surfaces are directly and mechanically connected to the connection surface of the second conductor plate, the third conductive surfaces are directly and mechanically connected to the connection surface of the

third conductor plate, the first conductive surfaces, the second conductive surfaces, and the third conductive surfaces are each flat for ensuring electrical and thermal conduction, the second conductor plate is contacted to the one terminal of the first rectifier element for electrical and thermal connection, the third conductor plate is contacted to the one terminal of the second rectifier element for electrical and thermal connection, wherein, the connection surface of the first conductor plate, the connection surface of the second conductor plate and the connection surface of the third conductor plate cover the whole common connection region, and the area of the connection surface of the first conductor plate is the largest, a cavity for flowing a cooling medium is provided inside the first conductor plate, the one terminal of the positive coil and the one terminal of the negative coil are continuously integrated through a portion provided with a conductive flat surface, the other terminal of the positive side coil protrudes from the central portion of the integrated conductive flat surface, the other terminal of the negative side coil protrudes from the central portion of the integrated conductive flat surface, the first unit has no unevenness plane on the integrated conductive flat surface, the second unit has one step higher plane both ends on the integrated conductive flat surface, the first unit and the second unit are alternately arranged and form the uneven surface of the first conductive surfaces, the first conductor plate includes an annular portion that covers and contacts a peripheral portion of the common connection region, and a cavity for circulating the cooling medium is provided inside the annular portion, the first conductive surfaces are arranged in a ring shape on the common connection region to surround the entire surface of the common connection region, and the connection surface of the first conductor plates is formed with the uneven surface include convex portions and concave portions, the connection surface is in close contact with the first conductive surfaces on the common connection region.

2. The welding transformer according to claim 1, wherein a connection surface of the second conductor and a connection surface of the third conductor are placed in a region surrounded by annular portion of the first conductor plate.

3. The welding transformer according to claim 1, wherein the connection surface of the second conductor plate is formed with an uneven surface including convex portions, and the connection surface of the second conductor is in close contact with the second conductive surfaces on the common connection region, and the connection surface of the third conductive plate is formed with an uneven surface including convex portions and concave portions, and the connection surface of the third conductive plate is in close contact with the third conductive surfaces on the common connection region.

4. The welding transformer according to claim 1, wherein the second conductor plate and the third conductor plate are provided with a cavity for circulating the cooling medium inside the second conductor plate and the third conductor plate.
